



CHAPTER 09

File I/O

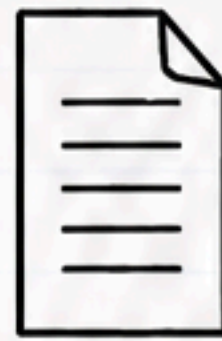


The random-access memory is **volatile**, and all its contents are lost once a program terminates. In order to **persist the data** forever, we use files.

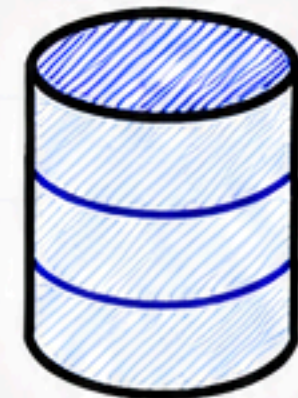
A **file** is data stored in a storage device. A python program can talk to the file by **reading** content from it and **writing** content to it.



Python Program



File
(Stored in Disk)



Storage Device
(Disk)



Type Of Files.

There are 2 types of files:

1. Text files (.txt, .c, etc)
2. Binary files (.jpg, .dat, etc)

1. Text files
(.txt, .c, etc)



- Human readable
- Stores text

2. Binary files
(.jpg, .dat, etc)



- Not human readable
- Stores images, data, etc.

Python has a lot of functions for reading, updating, and deleting files.



Opening A File.

Python has an **open()** function for opening files.

It takes 2 parameters: **filename** and **mode**.

`open("filename", "mode")`

Name of the file

Mode of opening
(read mode by default)

```
# open("filename", "mode of opening(read mode by default)")  
open("this.txt", "r")
```



Files make our programs
powerful and useful!





Reading A File In Python

```
# Open the file in read mode
f = open("this.txt", "r")
# Read its contents
text = f.read()
# Print its contents
print(text)
# Close the file
f.close()
```

How it works?

Open file
in read mode

Step 1



Read its
contents

Step 2



Print the
contents

Step 3

Close
the file

Step 4



Other Methods To Read The File.

We can also use `f.readline()` function to read one full line at a time.

```
f.readline() # Read one line from the file.
```



Reads one
line at a
time!

Modes Of Opening A File

Mode	Description
r	open for reading
w	open for writing
a	open for appending
+	open for updating.

'rb' will open for
read in binary mode.

'rt' will open for
read in text mode.



Files help us
store data
permanently!



Write Files In Python

In order to write to a file, we first open it in **write** or **append** mode after which, we use the python's **f.write()** method to write to the file!

```
# Open the file in write mode
f = open("this.txt", "w")

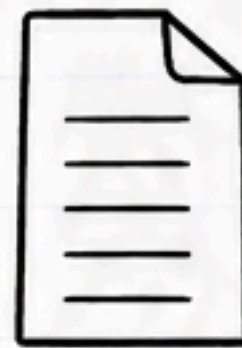
# Write a string to the file
f.write("this is nice")

# Close the file
f.close()
```



We can create a new file or overwrite the existing file using write mode.

this.txt



After Execution

this is nice



With Statement

The best way to open and close the file automatically is the **with** statement.

```
# Open the file in read mode using 'with'
with open("this.txt", "r") as f:
    # Read the contents of the file
    text = f.read()

# Print the contents
print(text)
```

How it works?

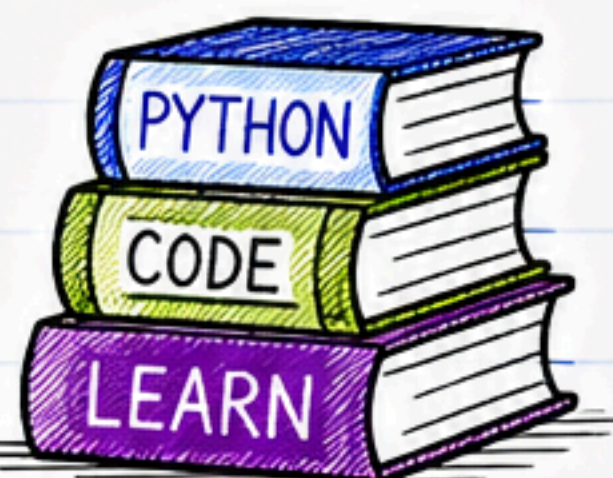
Open the file
(Automatically)

Perform operations
(read / write)

Close the file
(Automatically)

The file is
closed even
if an error
occurs!

Using with statement ensures better resource management and clean code. It automatically closes the file when the block is exited.

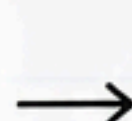
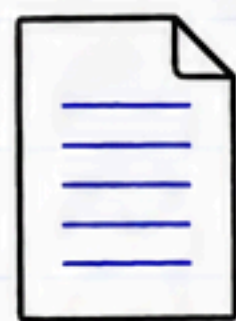




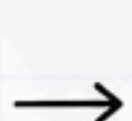
Practice Set

1. Write a program to read the text from a given file 'poems.txt' and find out whether it contains the word 'twinkle'.

poems.txt



Python Program



Contains 'twinkle'?

Yes

No

True

False



2. The game() function in a program lets a user play a game and returns the score as an integer. You need to read a file 'Hi-score.txt' which is either blank or contains the previous Hi-score. You need to write a program to update the Hi-score whenever the game() function breaks the Hi-score.

How it works?

User plays game



game() returns score



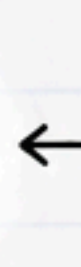
Score > Hi-score?

No

Hi-score unchanged

Yes

Hi-score.txt

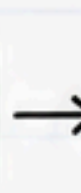


Update Hi-score in 'Hi-score.txt'



3. Write a program to generate multiplication tables from 2 to 20 and write it to the different files. Place these files in a folder for a 13-year-old.

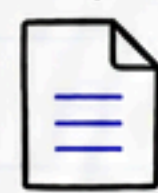
Python Program



Generates tables 2 to 20

(Each in separate file)

Folder

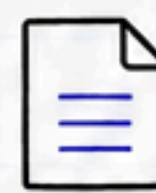


2.txt



3.txt

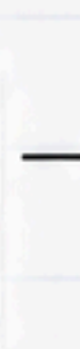
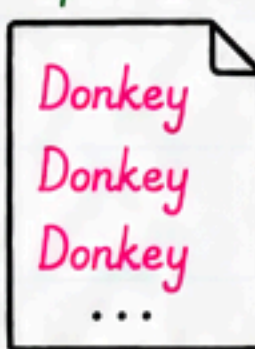
...



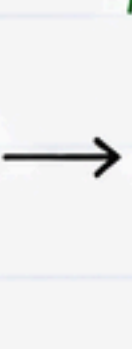
20.txt

4. A file contains a word "Donkey" multiple times. You need to write a program which replaces this word with ##### by updating the same file.

Input File



Python Program

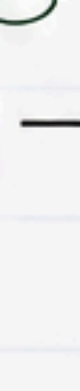


Updated Same File



5. Repeat program 4 for a list of such words to be censored.

List of words to be censored



Python Program



Updated Same File



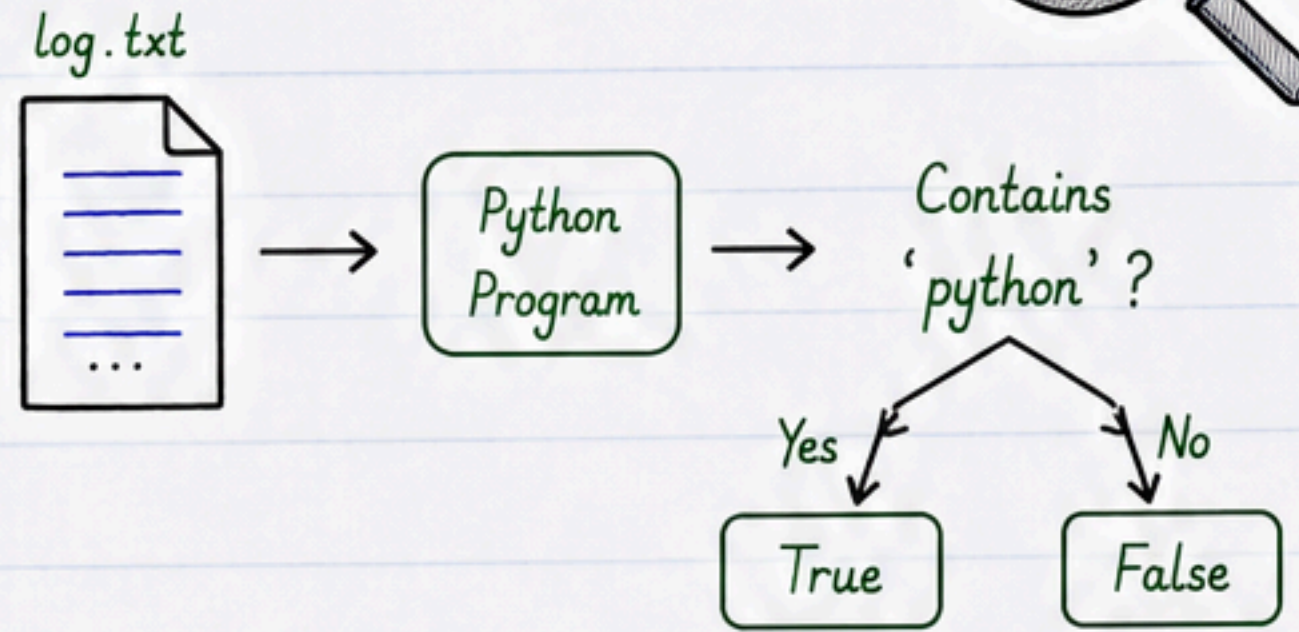
Practice makes perfect!
Keep coding, keep learning!



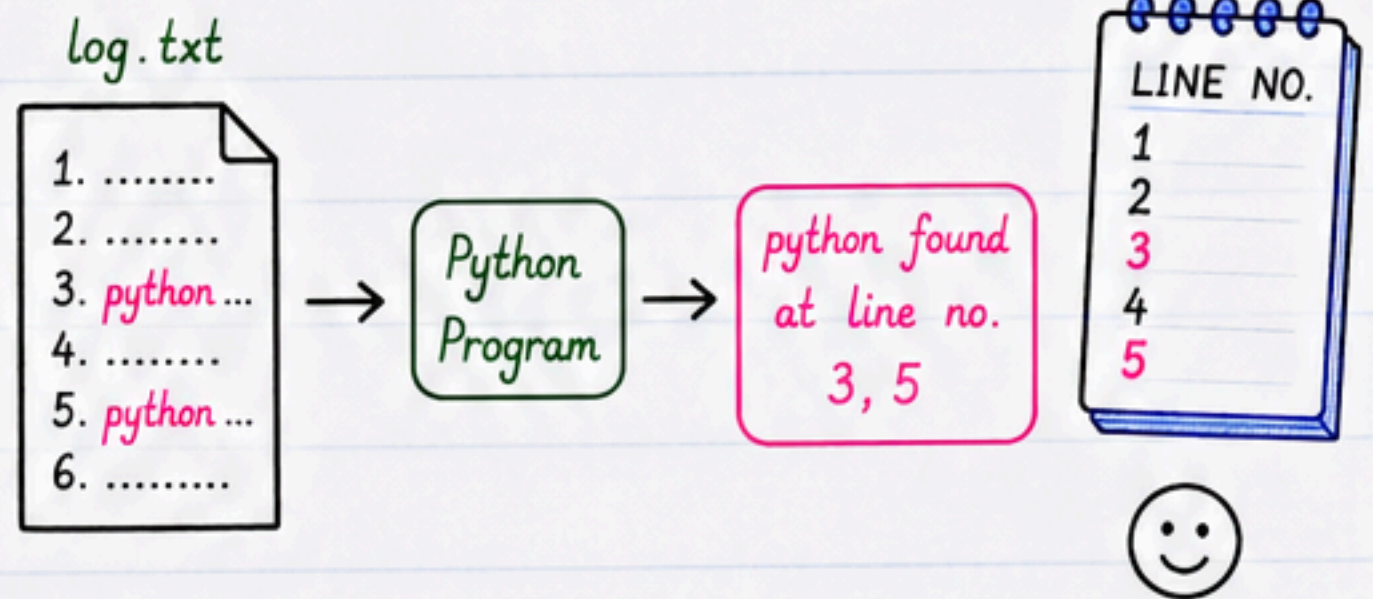
★ Practice Set (Continued)



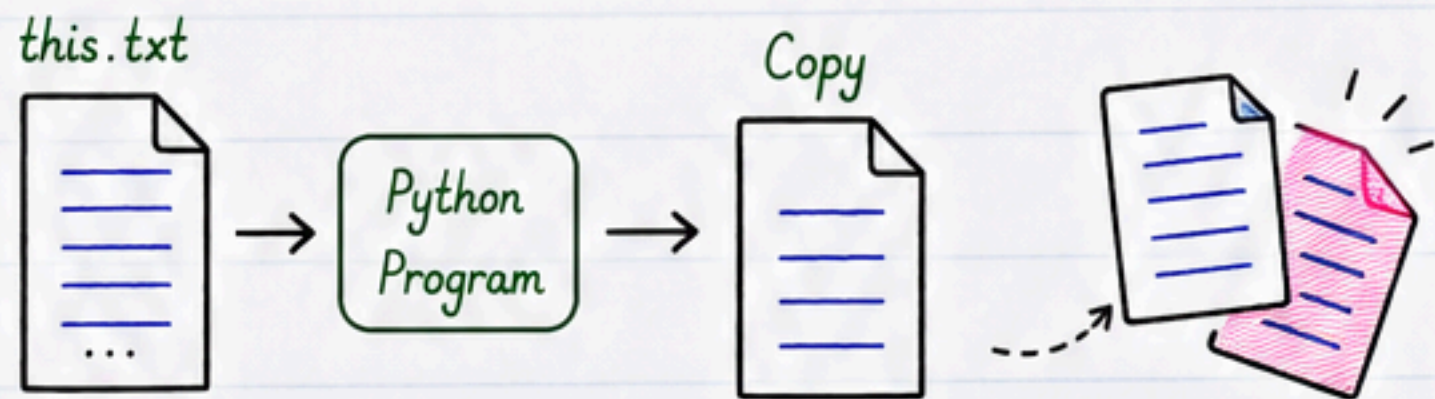
6. Write a program to mine a log file and find out whether it contains 'python'.



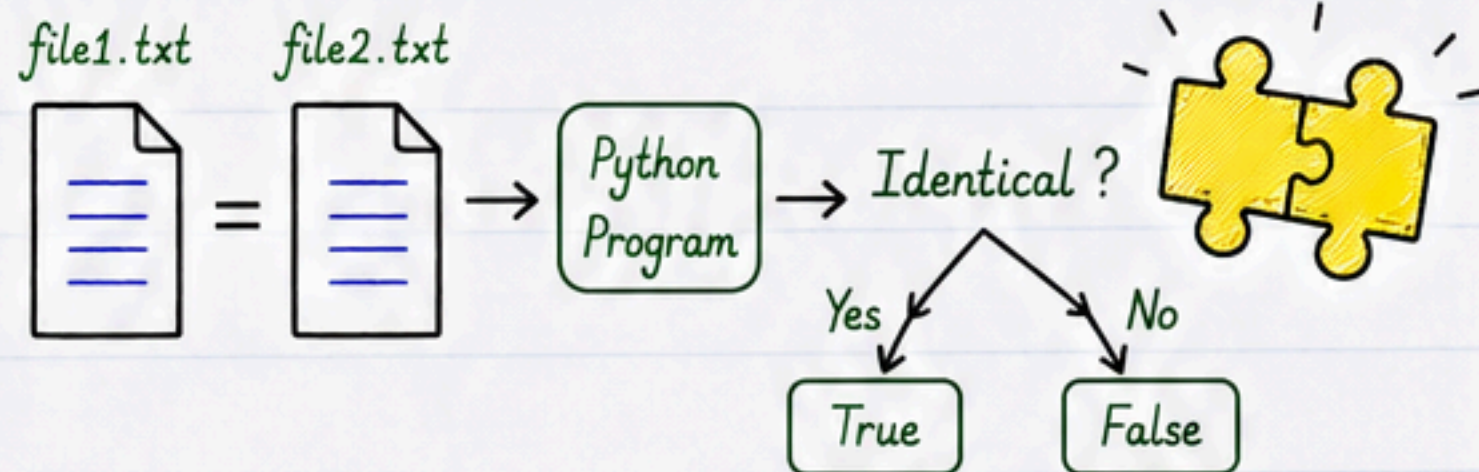
7. Write a program to find out the line number where python is present from ques 6.



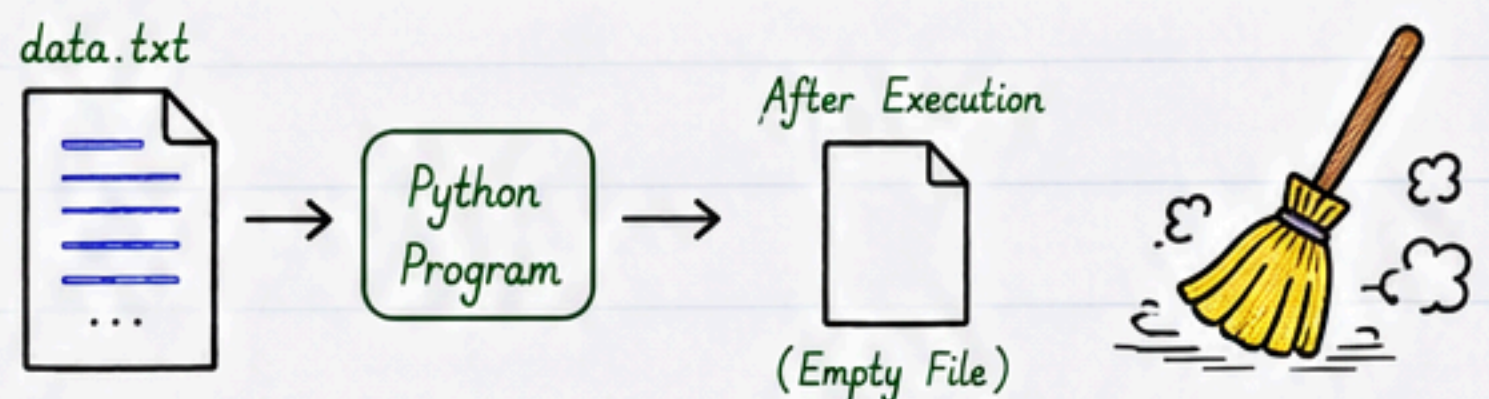
8. Write a program to make a copy of a text file "this.txt".



9. Write a program to find out whether a file is identical and matches the content of another file.



10. Write a program to wipe out the content of a file using python.



11. Write a python program to rename a file to 'renamed_by_python.txt'.

